

Business Loan Risk Model

Kestrel Bank — Credit Risk Analytics

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1. Ask Phase

Business Task

Written from the perspective of a loan analyst at Kestrel Bank, a UK business bank. The task: build a risk model that flags early signs of financial distress in a borrower, using UK public company records.

Problem

Banks usually act only after a missed payment. By then, the borrower may already be insolvent. Public records show warning signs earlier, such as officer resignations, late filings, and new charges.

Hypothesis

Companies that later fail show these signals more often, and earlier, than companies that stay active.

Metrics

Success is measured by how often the model correctly flags a company that later fails. Failure means a formal insolvency status. About 99% of the failed companies are in Liquidation or Administration, and the rest are in closely related procedures (receivership or a voluntary arrangement). “Dissolved” is excluded, since it also covers normal, healthy business closures.

Scope

The model covers Ltd, PLC, and LLP companies. Sole traders are excluded, since they aren’t registered with Companies House.

Stakeholders

Kestrel Bank’s credit risk team. Used for two decisions: approving new loan applications, and reviewing existing loans at mid-term audits. The model informs these decisions. It does not automate them.

2. Prepare Phase

Data Sources

Four datasets, all from Companies House, the UK's official company registrar. **The first** is the Free Company Data Product, a bulk download covering all 5.5 million registered UK companies, refreshed monthly. **The other three** — officer and resignation data, insolvency filing history, and registered charges — have no bulk download, so each is pulled company by company through Companies House's public API instead.

Location & Structure

Everything lives in BigQuery, The tables below are the starting point; others are added in later phases:

<i>raw_basic_company_data</i>	The full 5.5 million-row bulk file, every UK company on the register, kept as-is.
<i>sample_companies</i>	10,000 companies drawn from it: 5,000 in formal insolvency , 5,000 active , matched in equal numbers since real failures are too rare to learn from at their true, far smaller share.
<i>officers_output</i>	Director and company secretary appointment and resignation records for those same 10,000 companies, pulled from Companies House's API one company at a time.

The raw file stays untouched throughout, so the original data is always there to check back against.

Credibility & Licensing

All four datasets come directly from the UK government's official registrar, not a third party. Companies are legally required to keep this information current, and it's the same register banks and credit agencies rely on themselves. The data is published under the UK's Open Government Licence, free to reuse, and officer details are public by law — anyone who's a company director has their name and role on public record, not something gathered without their knowledge.

Integrity Checks

Before filtering, every real status value in the bulk file was checked by hand rather than assumed, since Companies House doesn't publish a fixed list. Row counts were confirmed at each step: exactly 10,000 companies sampled, an officer-data API pull attempted for all of them, and known companies (e.g. Tesco, Wilko) spot-checked to confirm the data reads correctly.

Limitations

The bulk file covers every registered entity type, not just Ltd, PLC, and LLP — it also includes Limited Partnerships and Overseas Entities, which fall outside this project's scope and need filtering out. Officer history is also sometimes incomplete for companies that have been insolvent for a long time, likely archived once the process closes.

3. Process Phase

Tools

SQL in BigQuery for the bulk company data. The officer and resignation records were acquired directly from Companies House, the same source as everything else.

Loading the Data

The bulk file (2.61GB) was loaded into BigQuery, since it's too large for Sheets or Excel to open. Two real issues came up:

- A column name containing a period (RegAddress.CareOf), fixed via BigQuery's column-name character map setting;
- and a small number of malformed rows, handled by raising the load's allowed-errors threshold instead of rejecting the whole file.

Filtering to the Relevant Companies

This is the main cleaning step: instead of working with all 5.5 million companies, the data is narrowed down to just the ones relevant to the model. So few companies actually end up in formal insolvency, that using the full dataset would give the model very little to learn from — a smaller, balanced sample, the same number of struggling companies as healthy ones, is the standard approach instead.

The filtering was done in three simple steps.

Step 1 — a random sample of companies in formal insolvency:

```
SELECT *  
FROM raw_basic_company_data  
WHERE TRIM(CompanyStatus) IN ('Liquidation', 'In Administration', ...)  
ORDER BY RAND()  
LIMIT 5000
```

(the real list has eight status values, mostly Liquidation and In Administration variants; shortened here for readability). `TRIM()` strips any accidental extra spaces around the text, so a value like 'Active' with a trailing space still matches cleanly. `WHERE` keeps only the rows whose status means real financial distress. `ORDER BY RAND()` shuffles those rows randomly, so `LIMIT 5000` picks a random 5,000.

Step 2 — the same number of healthy/ active companies, for a fair comparison:

```
SELECT *  
FROM raw_basic_company_data  
WHERE TRIM(CompanyStatus) = 'Active'  
ORDER BY RAND()  
LIMIT 5000
```

Step 3 — combine both into one table of 10,000. `UNION ALL` two sets of rows into one list, while

`CREATE OR REPLACE TABLE` saves a query's result as an actual table — both are needed together here:

```
CREATE OR REPLACE TABLE
`skilful-answer-498915-k5.Project_1.sample_companies` AS
WITH distressed AS (
  SELECT *
  FROM raw_basic_company_data
  WHERE TRIM(CompanyStatus) IN ('Liquidation', 'In Administration', ...)
  ORDER BY RAND()
  LIMIT 5000
),
active AS (
  SELECT *
  FROM raw_basic_company_data
  WHERE TRIM(CompanyStatus) = 'Active'
  ORDER BY RAND()
  LIMIT 5000
)
SELECT * FROM distressed
UNION ALL
SELECT * FROM active;
```

`WITH ... AS` just gives each of the two smaller queries above a name (`distressed`, `active`). `UNION ALL` stacks their results together, and `CREATE OR REPLACE TABLE` saves that combined result as `sample_companies`.

Building the Officer, Filing, and Charges Data

None of these three have a bulk download, so each was pulled the same way: one API request per company, for all 9,844 companies.

Officers	Director and company secretary appointments and resignations. Signal: whether people started leaving before things went wrong.
Filing history	Insolvency filings (administration orders, winding-up resolutions). Gives a real, dated failure event, not just a current-status snapshot.
Charges	Registered claims against a company's assets, usually securing a loan. Signal: how much is at risk.

Checking the Officer data

Checking officer records — director and company secretary appointments — against the 10,000-company sample found 614 with none at all. 156 of those are non-standard entity types (Limited Partnerships and similar) — structured around partners rather than directors, so they fall outside this model's scope; 153 make up part of the 614, and the other 3 had officer records anyway, just an inconsistency in Companies House's data. The remaining 461 are legitimate companies with genuinely missing history, likely archived by Companies House once a company's been closed a long time — a limitation, not something to fix.

```
DELETE FROM `skilful-answer-498915-k5.Project_1.sample_companies`  
WHERE SUBSTR(CompanyName, 1, 2) IN ('LP', 'SL', 'OE', 'CE', 'CS',  
'IP', 'RS', 'NF', 'SG')
```

Matches the first two letters of each company number against those non-standard types (Limited Partnerships and similar) and removes them. Brings the sample from 10,000 to 9,844; since all 156 are on the active side, that side drops to 4,844 — no longer a perfect 50/50 split.

Checking the Filing-History and Charges Data

The first pull capped results at 100 records per company, so the 8 companies with the most activity got cut short — Wye Finance Co Limited was the worst case, with only 100 of its 325 real charges captured. A second pass re-fetched just those 8 with the fix applied. Final counts: 9,963 rows in **charges_output**, 19,780 in **filing_history_output**.

Data Cleaning

Before moving to analysis, the combined dataset went through a standard set of checks:

- **Duplicates** — none; all 9,844 companies have a unique CompanyNumber.
- **Whitespace** — a stray leading space in the original "CompanyName" header was stripped before it could break joins and lookups.
- **Casing** — CompanyStatus values are inconsistent (e.g. "RECEIVERSHIP" vs "In Administration"); the sampling query lists every variant and wraps comparisons in TRIM().
- **Primary key** — CompanyNumber, used across all tables.
- **Data types** — CompanyNumber loaded as STRING (leading zeros intact); dates loaded as proper dates, confirmed against BigQuery's schema.
- **Dates** — already ISO YYYY-MM-DD in both the bulk file and the API, no conversion needed.
- **Currency** — GBP only, no conversion needed.

Building the Company Risk Features Table

One row per company, combining the three signals pulled so far. For failed companies, the reference point is the earliest insolvency filing on record — the real start of that company's failure, not just its current status. For active companies, it's the snapshot date, so every company is measured over the same 12-month window even though the anchor date means something different for each. From there: director resignations in that window, whether a filing was overdue at that point, and total and outstanding charges as of that date.

Step 1 — Setting the reference date

Sets each company's reference date — the point in time everything else gets measured "as of." For failed companies that's their earliest insolvency filing; if there isn't one (including every active company), it falls back to the snapshot date. Without this, you'd be comparing a failed company's final months against an active company's status today, which isn't a fair comparison:

```
CREATE OR REPLACE TABLE
`skilful-answer-498915-k5.Project_1.company_risk_features` AS
WITH reference_dates AS (
  SELECT
    s.CompanyNumber AS company_number,
    s.CompanyName AS company_name,
    s.outcome_label,
    s.Accounts_NextDueDate,
    s.ConfStmntNextDueDate,
    COALESCE(fh.first_filing_date, DATE('2026-09-01')) AS reference_date
  FROM `skilful-answer-498915-k5.Project_1.sample_companies` s
  LEFT JOIN (
    SELECT company_number, MIN(filing_date) AS first_filing_date
    FROM `skilful-answer-498915-k5.Project_1.filing_history_output`
    GROUP BY company_number
  ) fh ON fh.company_number = s.CompanyNumber
),
```

Step 2 — Counting director resignations

Counts director resignations in the 12 months before each company's reference date. A cluster of resignations right before failure is a known early-warning signal, so this turns the raw officer records into one number per company:

```
resignations AS (
  SELECT r.company_number, COUNT(o.resigned_on) AS
director_resigned_in_window
  FROM reference_dates r
  LEFT JOIN `skilful-answer-498915-k5.Project_1.officers_output` o
    ON o.company_number = r.company_number
    AND o.resigned_on BETWEEN DATE_SUB(r.reference_date, INTERVAL 12
MONTH) AND r.reference_date
  GROUP BY r.company_number
),
```

Step 3 — Counting charges

Counts each company's charges (loans/collateral secured against the business) as of its reference date — total registered, and how many were still outstanding at that point. A build-up of unresolved charges is a structural sign of financial strain:

```
charges AS (  
  SELECT  
    r.company_number,  
    COUNTIF(c.created_on <= r.reference_date) AS  
total_charges_as_of_ref,  
    COUNTIF(c.created_on <= r.reference_date AND (c.satisfied_on IS NULL  
OR c.satisfied_on > r.reference_date)) AS outstanding_charges_as_of_ref  
  FROM reference_dates r  
  LEFT JOIN `skilful-answer-498915-k5.Project_1.charges_output` c  
    ON c.company_number = r.company_number  
  GROUP BY r.company_number  
)
```

Step 4 — Combining everything

Brings the three pieces together into one row per company, using IFNULL so a company with no matching resignation/charges rows gets a clean 0 (or FALSE) instead of NULL, and combines the two due-date fields into a single overdue flag:

```
SELECT  
  r.company_number,  
  r.company_name,  
  r.outcome_label,  
  r.reference_date,  
  IFNULL(res.director_resigned_in_window, 0) AS  
director_resigned_in_window,  
  (IFNULL(r.Accounts_NextDueDate < r.reference_date, FALSE)  
  OR IFNULL(r.ConfStmntNextDueDate < r.reference_date, FALSE)) AS  
filing_overdue_at_ref,  
  IFNULL(ch.total_charges_as_of_ref, 0) AS total_charges_as_of_ref,  
  IFNULL(ch.outstanding_charges_as_of_ref, 0) AS  
outstanding_charges_as_of_ref  
FROM reference_dates r  
LEFT JOIN resignations res ON res.company_number = r.company_number  
LEFT JOIN charges ch ON ch.company_number = r.company_number;
```

Checking the Company Risk Features Table

9,844 rows, matching the sample exactly. Failed companies score higher than active ones on every signal:

	Active	Failed
Avg. resignations (12mo)	0.13	0.29
Overdue at reference date	5.9%	47.4%
Avg. total charges	0.64	1.35
Avg. outstanding charges	0.33	0.74

One limitation worth noting: 795 of the 5,000 failed companies (16%) began failing in the 1980s or 90s, before Companies House's digital officer and charges records existed, so they show no measurable activity in these windows — a data-availability gap, not a modeling error. The result holds up despite it: failed companies still clearly outscore active ones across every signal.

4. Analyse Phase

Backtest Results

To test whether these signals would actually be useful for flagging risk, the four features get collapsed into a single 0–3 risk score — one point each for at least one director resignation in the window, an overdue filing at the reference date, and at least one outstanding charge. Total charges are left out of the score itself since it overlaps heavily with outstanding charges: a company with several old, fully-repaid charges isn't riskier than one with none, only unresolved ones are the relevant signal. Each company's score is then tested against four thresholds to see what fraction of failed companies would have been caught, versus what fraction of active companies would have been wrongly flagged.

Step 1 — Scoring each company

Turns the three signals into a single score per company:

```
WITH scored AS (  
  SELECT  
    company_number,  
    outcome_label,  
    IF(director_resigned_in_window >= 1, 1, 0)  
      + IF(filing_overdue_at_ref, 1, 0)  
      + IF(outstanding_charges_as_of_ref >= 1, 1, 0) AS risk_score  
  FROM `skilful-answer-498915-k5.Project_1.company_risk_features`  
)
```

Step 2 — Measuring detection vs. false positives by threshold

For each possible threshold, checks what share of failed companies would have been flagged (detection rate) against what share of active companies would have been wrongly flagged (false-positive rate):

```
SELECT
  threshold,
  ROUND(COUNTIF(outcome_label = 'failed' AND risk_score >= threshold) /
COUNTIF(outcome_label = 'failed'), 3) AS detection_rate,
  ROUND(COUNTIF(outcome_label = 'active' AND risk_score >= threshold) /
COUNTIF(outcome_label = 'active'), 3) AS false_positive_rate
FROM scored, UNNEST([0,1,2,3]) AS threshold
GROUP BY threshold
ORDER BY threshold;
```

Flagging on any single warning sign (score ≥ 1) catches 69.6% of failed companies, while wrongly flagging 24.2% of active ones — a real but manageable cost given how much it catches. Raising the bar to two or more signals sharply cuts false positives but also detection, catching only 20.1% of failures; requiring all three catches just 2.1%.

Threshold	Detection rate	False-positive rate
≥ 0	100%	100%
≥ 1	69.6%	24.2%
≥ 2	20.1%	2.3%
≥ 3	2.1%	0.1%

One limitation worth noting: this is measured on a roughly balanced sample of failed and active companies, not the real-world failure rate, which is far lower in any given year. So these numbers show how well the score discriminates between failed and active companies in this test set — they shouldn't be read as "X% of flagged companies will actually fail," since that depends on the true proportion of at-risk companies in a real loan book.

5. Share Phase

Audience & Formats

Kestrel Bank's credit risk team, who use the model for approving new loans and reviewing existing ones at mid-term audits. The findings are shared in three formats, so each reader can go as deep as they need: a one-page executive brief, a slide deck, and an interactive Tableau dashboard.

Key Findings

Failed companies show warning signs far more often. 75.8% of active companies have no warning signs, against 30.4% of failed ones. The gap widens with each additional sign:

Warning signs	Active	Failed
0	75.8%	30.4%
1	21.9%	49.5%
2	2.2%	18.0%
3	0.1%	2.1%

Overdue filings are the clearest single signal. 47.4% of failed companies had a filing overdue at the reference date, against 5.9% of active ones, roughly 8 times higher.

Each extra warning sign trades detection for fewer false alarms. See the threshold table in the Analyse phase. One sign catches 69.6% of failures, while three isolate only the clearest cases.

Reading the Results

The sample is balanced between failed and active companies, so the results show how well the score separates the two groups, not the real-world chance that a flagged company fails.

6. Act Phase

Conclusion

Three public-record signals (director resignations, overdue filings, outstanding charges) combine into a simple 0–3 score that separates failing companies from healthy ones. Flagging on one sign catches 69.6% of failures, and each additional sign sharply cuts false alarms.

Recommendations

Warning signs	Action
1	Flag for closer review before loan approval, not automatic rejection
2	Escalate to a mid-term audit of the existing loan
3	Escalate to a senior underwriter for rejection review

The model informs lending decisions. It does not automate them.

Limitations

- The balanced sample shows relative risk, not real-world default rates.
- 16% of failed companies (795 of 5,000) failed before digital records existed.
- Sole traders and other non-registered businesses are out of scope.
- There is no fixed threshold. The right cut-off depends on Kestrel's risk appetite.

Next Steps & Further Exploration

- **Loan exposure estimate:** estimate potential loss per flagged company, scaled by company size band (British Business Bank, 2025).
- **Real-world calibration:** re-run on a sample that reflects actual failure rates, to give true probabilities.
- **Lead time:** measure how many months before failure each signal first appears.
- **Pilot:** test the score on a live portion of the loan book and set thresholds with the credit risk team.